Chapter 7 Forces

EXTRA WORKSHEET 2: Safer landings

SCIENCE AS A HUMAN ENDEAVOUR **VERBAL/LINGUISTIC**

Name:_____Class:_____Date: _____

Sand playground makes for safer landings

ANNABEL MCGILVRAY

In the often high-tech world of modern children's playgrounds, a new study suggests that an old-fashioned solution may be best.

Canadian researchers have found that children are nearly five times less at risk of breaking arms, legs or parts of their bodies, if their playground has sand on the ground rather than a wood fibre base.

The Toronto team's results are published on the PLoS Medicine website today and may send local councils and planning authorities across Australia scrambling back to their drawing boards.

'We suspect that the fracture rates are lower on sand because of lower surface friction,' write the Toronto-based researchers. 'A lower friction surface allows the hand to slide or sink limiting the bending moment and preventing a fracture.'

'Updating playground safety standards to reflect this information will reduce the most common and severe injuries seen on modern playgrounds, without limiting children's access to healthy outdoor play.'

Breaking new ground

The researchers describe how they took advantage of a 2003 decision by the Toronto District School Board to resurface the district's school playgrounds. They used the rare opportunity to conduct a randomised controlled trial of the two surfaces.

It found that children playing in those playgrounds with wood fibre surfaces were 4.9 times more likely to suffer fractures than the children playing in sand-covered playgrounds. The rate of fractures reported for the sandy playgrounds was 1.9 for every 100 000 student-months of use.

Australian playground safety expert and associate professor at the University of Technology, Sydney, David Eager says that it is the first such study comparing surfaces in real world conditions and it is very thorough.

'It's an excellent study and long-overdue.'

In Australia, a recent study of playground safety found that over a ten-year period there were 106.6 fractures per 100 000 students.

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Eager attributes the Australian injury rate in part to the fact that many playgrounds around the country are now surfaced with rubber. His recent research has shown that rather than dispersing the energy generated by a fall, rubber rebounds energy back into the falling body.

'My own opinion is that both sand and wood fibre, or bark, are excellent surfaces for playgrounds and it's the rubber that is the bad surface which causes the rebound and lots more breaks.'

'The forces are significantly greater on the rubber than they are on the bark or the sand.'

Sinking to safe depths

Adequate maintenance of playground surfaces can also play an important role in safety and may have influenced the impressively small injury outcome in the Canadian study.

Any coverings of depth, such as sand or wood fibre, should be maintained at optimum depth to have the desired effect on safety.

While no measurements were made, the researchers point out that the playgrounds were well looked after. They add that not only were the surfaces newly laid, but that the supervisors of the participating playgrounds were aware of the research and worked to maintain the grounds.

Previous research by the NSW Injury Risk Management Research Centre has demonstrated that the safety benefit of such surfaces is significantly diminished if they are not as deep as required. The Centre suggested that it is one area often overlooked during maintenance.

'In Australia we have lots of old playgrounds and lots of playgrounds that aren't maintained,' says Eager.

'When we do have sand and bark, even if it is put in at the correct depth, at the bottom of slides and climbing poles and other forced movement devices, it all gets pushed away and you're left with something quite hard underneath which is probably where all the injuries are occurring.'

Source: ABC Science website

1 Researchers from Canada found that children who played in playgrounds with a base of sand were at a much lower risk of breaking a bone in a fall than those who played in playgrounds with a wood fibre base. Recall how less likely a break is, according to the article.

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2 State the reason why the researchers believe that this is the case.

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3 Explain why having an arm or leg being able to slide in the sand would reduce the chance of a break.

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4 State the number of fractures that was found over a ten year period out of a total of 100 000 children.

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5 Safety expert David Eager criticised the use of rubber surfaces in many Australian playgrounds. Outline his reasons for this view.

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6 Explain why it is important that playgrounds are well maintained to prevent injuries.

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